

Multiple Choice (1 point each)

Question 1

Heap is a suitable data structure for building priority queues.

- ☐ True
 - ☐ False
-

Question 2

Heap can be built in linear time by repetitive insertion of data into the heap.

- ☐ True
 - ☐ False
-

Question 3

The worst-case time complexity of the naïve pattern matching algorithm is:

$$O(mn)$$

where m is the length of the pattern and n is the length of the text.

- ☐ True
 - ☐ False
-

Question 4

Counting Sort is a comparison-based algorithm.

- ☐ True
 - ☐ False
-

Question 5

In Radix Sort, we focus on digits or keys of the data set.

- ☐ True
☐ False
-

Question 6

In the KMP algorithm, when a mismatch occurs, the pattern itself provides enough information to determine where the next match begins.

- ☐ True
☐ False
-

Question 7

In tries, the number of comparisons required while searching for a word is equal to the length of the word.

- ☐ True
☐ False
-

Question 8 (3 points)

There may be multiple correct answers. Mark all correct combinations.

Suppose:

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

Then:

- (i) $\Theta(n^{\log_b a})$
- (ii) $\Theta(n^{\log_b a} \log n)$
- (iii) $\Theta(f(n))$

Under the following conditions:

- (a) $f(n) = \Theta(n^{\log_b a})$
- (b) $f(n) = O(n^{\log_b a - \varepsilon})$, $\varepsilon > 0$
- (c) $f(n) = \Omega(n^{\log_b a + \varepsilon})$ and $af(n/b) \leq cf(n)$, $c < 1$

- ☐ (ii), (a)
- ☐ (ii), (b)
- ☐ (iii), (c)
- ☐ (i), (c)
- ☐ (iii), (a)
- ☐ (i), (b)